

Out-of-Sample Prediction of the Finite Shape Clock

Affine, Logistic, and Power-Gap Laws on Two Physical Channels

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July 2026

Abstract

The sixteen-level audit of RH-214 finds a strictly increasing axial quartet coordinate u and a narrow mature asymmetry corridor. This paper asks whether that finite clock supports a simple predictive law. Before scoring the finest two levels, we fit three predeclared models on seven scales from $\sigma = 0.008$ through 0.002: an affine law in $\log(1/\sigma)$, an affine logit law, and a power law for the gap $1 - u$.

All three fit the training segment well. On the left channel their direct u -scale training R^2 values are 0.99266, 0.99272, and 0.98716. The held-out levels 0.0016 and 0.00125 reverse the in-sample ranking. The power-gap model has two-point RMS error 0.00989 on the left and 0.01003 on the right; logistic errors are about 0.0206 and affine errors about 0.0305. Power-gap wins each held-out point on each channel, but it still overpredicts u by roughly 0.01.

A constant predictor for mature η misses the holdouts by at most 0.00218, consistent with a corridor but not a limit. The strict conclusion is a finite model-selection result. No asymptotic power law, value $\lim u = 1$, or determinant limit is established.

1 Prediction rather than retrospective fit

RH-214 shows that u is ordered across sixteen scales [1]. A smooth curve can fit such a finite monotone list almost arbitrarily well, so in-sample residuals alone have little force. We instead freeze

$$\sigma \in \{0.008, 0.00625, 0.005, 0.004, 0.0032, 0.0025, 0.002\} \quad (1)$$

for training and reserve

$$\sigma \in \{0.0016, 0.00125\} \quad (2)$$

for scoring. Both channels use the same split and model classes.

This is a deterministic numerical experiment, not an independent random sample. “Out of sample” means only that the held-out coordinates are not used in estimating the reported coefficients.

2 Three predeclared laws

Let

$$t = \log(1/\sigma). \quad (3)$$

For each channel we estimate an ordinary least-squares line after one of three transforms.

2.1 Affine-log model

$$u(t) = at + b. \quad (4)$$

This is the least constrained local model. It cannot be globally valid with $a > 0$, because it eventually exceeds one.

2.2 Logistic-log model

$$\log \frac{u(t)}{1 - u(t)} = at + b, \quad u(t) = \frac{1}{1 + e^{-at-b}}. \quad (5)$$

For $a > 0$, it enforces $0 < u < 1$ and approaches one.

2.3 Power-gap model

$$\log(1 - u(t)) = at + b, \quad 1 - u = e^b \sigma^{-a}. \quad (6)$$

The fitted a is negative, so $-a$ is the candidate gap exponent.

The logistic model also has a power-gap leading asymptotic when its logit is large, but the two forms weight the finite training region differently.

3 Training fit

For the left channel:

model	direct- u R^2	max training error	RMS training error
affine-log	0.992663	0.008231	0.005895
logistic-log	0.992719	0.008117	0.005872
power-gap	0.987164	0.012233	0.007797

The right channel gives the same qualitative ranking, with R^2 between 0.98725 and 0.99285. If the analysis stopped here, one would select affine or logistic and might infer an overly rapid advance.

The fitted left formulas are approximately

$$u_{\text{aff}}(t) = 0.149086t - 0.242734, \quad (7)$$

$$\text{logit } u_{\text{log}}(t) = 0.619923t - 3.091628, \quad (8)$$

$$\log(1 - u_{\text{gap}}(t)) = -0.361228t + 1.112179. \quad (9)$$

4 Held-out verdict

The actual left coordinates are

$$u(0.0016) = 0.692811, qquad u(0.00125) = 0.718352. \quad (10)$$

The predictions are:

model	prediction at 0.0016	prediction at 0.00125	two-point RMS
affine-log	0.717047	0.753850	0.030393
logistic-log	0.710802	0.741220	0.020574
power-gap	0.702790	0.728146	0.009886

On the right, the actual values are 0.692310 and 0.717501. The corresponding RMS errors are 0.030639, 0.020825, and 0.010031.

Proposition 4.1 (Finite holdout ranking). *On both physical channels, the power-gap model has the smallest absolute error at each of the two held-out scales and the smallest two-point RMS error among the three predeclared models.*

Proof. The assertion is the direct numerical comparison of the frozen predictions and held-out coordinates above. The right-channel inequalities are archived with the same ordering. $\square$

The power-gap model nevertheless overpredicts both left holdouts by 0.00998 and 0.00979, and both right holdouts by about 0.010. Its win is relative, not an adequate asymptotic residual bound.

5 What the holdout teaches

The ordering of training and holdout errors contains more information than the winning label:

1. the affine and logistic curves extrapolate the recent rise too aggressively;
2. the power-gap transform tolerates more curvature in the observed finite window;
3. high R^2 on seven correlated deterministic anchors does not select a tail law;
4. both channels fail in nearly the same direction and magnitude, so the effect is unlikely to be explained solely by channel mismatch.

It would be inappropriate to attach conventional independent-sample p -values to these points. The relevant next evidence would be further predeclared scales or an analytic estimate from the underlying operator.

6 Transverse holdout

The training means of η are

$$\bar{\eta}_L = -0.090859, qquad \bar{\eta}_R = -0.092405. \quad (11)$$

Using these constants for both held-out levels gives maximum errors 0.002171 and 0.002039. This supports RH-214’s finite corridor language. It does not distinguish convergence to a constant from bounded slow drift or oscillation.

7 Conditional asymptotic readings

If (6) held for all sufficiently small σ with $a < 0$, then

$$1 - u_\sigma \sim C\sigma^\beta, \quad \beta = -a > 0, \quad (12)$$

and hence $u_\sigma \rightarrow 1$. The fitted exponents are approximately 0.3612 on both channels. This implication is mathematically correct but its premise is unproved. We therefore use the power-gap law only as a finite candidate coordinate for subsequent falsification.

RH-216 takes a different route: it proves exactly what $u \rightarrow 1$ would imply for the quartet roots and discriminant, without asserting that the physical sequence satisfies the premise.

8 Claim boundary

This paper establishes no asymptotic regression consistency, no error bars uniform in σ , and no value of $\lim u$. Two held-out points cannot differentiate a true power law from a slowly varying correction, a crossover, or another smooth monotone curve.

The strict result is the protocol and finite ranking in Proposition 4.1. Gate A remains open. No growing determinant, Hilbert–Pólya operator, zeta-zero identification, or RH implication is claimed.

References

- [1] Liang Wang. A finite monotone axial clock for the quartet shape flow, 2026. RH-214 preprint.